

## Internship Project Proposal

*Intelligence-Driven Purple Team Pipeline for E-Banking  
Infrastructure*

Submitted By:

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# INTERNSHIP PROJECT PROPOSAL

## *Intelligence-Driven Purple Team Pipeline for E-Banking Infrastructure*

Submitted to: **Prof. Rath**

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Year 3 Cybersecurity Student

Date: March 2026

|                   |                                                                       |
|-------------------|-----------------------------------------------------------------------|
| Project Title     | Intelligence-Driven Purple Team Pipeline for E-Banking Infrastructure |
| Student           | THY DAYUTH                                                            |
| Supervisor        | Professor Rath (Cryptography / Red & Blue Team Specialist)            |
| Host Organisation | ACLEDA Bank PLC — Cybersecurity Division                              |
| Duration          | Approximately 3 Months (Internship Period)                            |
| Classification    | Confidential — Internal Review Only                                   |

## 1. Executive Summary

Advanced financial threat actors — including nation-state groups and organised cybercriminal syndicates — continue to target banking infrastructure at an accelerating pace. Traditional Security Operations Centre (SOC) models, which rely primarily on signature-based detection and reactive alert triage, are increasingly insufficient against adversaries who operate with patience, precision, and a deep understanding of defensive tooling. As a result, institutions face a widening detection gap between the techniques employed by attackers and the capabilities available to defenders at the Tier 1 monitoring level.

This proposal outlines an Intelligence-Driven Purple Team Pipeline — a structured, four-phase internship project that moves the bank from reactive monitoring to proactive threat emulation and high-fidelity detection engineering. By selecting a real-world threat group that targets financial institutions, mapping its tactics to the MITRE ATT&CK framework, safely emulating its attack chain in a controlled environment, and then engineering precision SIEM detections and an Incident Response (IR) playbook, this project will produce three immediately deployable security artefacts. Every phase is deliberately aligned to the internship curriculum and designed to yield measurable improvements to the bank's security posture, compliance readiness, and SOC operational maturity.

## 2. Project Background & Problem Statement

### 2.1 The Evolving Threat Landscape

Financial institutions are among the most targeted sectors globally. According to IBM's Cost of a Data Breach Report, the financial industry consistently records the second-highest average breach cost across all industries, with sophisticated, multi-stage attacks frequently bypassing perimeter defences undetected for weeks or months. Threat groups such as Lazarus Group (linked to state-sponsored activity) and FIN7 (a financially motivated criminal syndicate) have demonstrated an ability to compromise banking networks through precisely crafted spear-phishing campaigns, living-off-the-land lateral movement, and targeted exfiltration of core banking databases.

Despite significant investment in security tooling, the root cause of persistent breaches is not a lack of data — it is a lack of validated, high-fidelity detections. Most SIEM rule sets are built from generic threat intelligence rather than from tested, adversary-specific telemetry. This means that even when an attacker executes a known technique, it may go undetected because no rule has ever been tested against that specific behaviour in the bank's unique environment.

### 2.2 Why This Problem Must Be Addressed Now

- Regulatory pressure is intensifying. Financial regulators increasingly mandate demonstrable threat-testing capabilities and documented IR playbooks.
- Internship curricula cover all required skill areas. The current training schedule — spanning Threat Intelligence, Penetration Testing, SIEM, Incident Response, and E-Banking Security — maps directly to this project's phases, making it the optimal time to integrate theory with practice.
- The Purple Team methodology is industry-recognised but rarely practised at internship level. Delivering a functional pipeline would represent a genuine contribution to the bank's security operations, not merely a student exercise.

## 3. Project Objectives (SMART Goals)

The following objectives define what success looks like at the conclusion of the internship period:

| SMART | Criterion | Objective | Target | Timeframe |
|-------|-----------|-----------|--------|-----------|
|-------|-----------|-----------|--------|-----------|

|          |            |                                                                                                                     |                                                  |                          |
|----------|------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|--------------------------|
| <b>S</b> | Specific   | Map one complete APT attack chain (Initial Access → Lateral Movement → DB Exfil) to MITRE ATT&CK                    | One fully documented threat actor profile        | End of Week 2 (Phase 1)  |
| <b>M</b> | Measurable | Produce SIEM detection rules with a false-positive rate below 5% in the test environment                            | 3+ custom detection rules deployed and tested    | End of Phase 3           |
| <b>A</b> | Achievable | Constrain scope to a single, well-documented attack vector executable with open-source tooling in a lab environment | Lab environment approval obtained before Phase 2 | Week 3                   |
| <b>R</b> | Relevant   | All phases directly address courses in the approved curriculum and the bank's real e-banking security needs         | 100% curriculum alignment across all four phases | Ongoing                  |
| <b>T</b> | Time-Bound | Deliver all three final artefacts (Runbook, SIEM Rules, IR Playbook) before the final internship presentation       | Three completed, documented artefacts            | Final week of internship |

## 4. Project Methodology & Approach

### 4.1 Scope Definition

To prevent scope creep and ensure a quality outcome within the internship timeframe, the project will focus on a single, end-to-end attack vector representative of threats faced by financial institutions:

**Spear-phishing Initial Access → Credential Harvesting → Lateral Movement → E-Banking Database Access**

What is explicitly OUT of scope:

- Production system testing — all emulation occurs in an isolated lab or approved development environment only.

- Full red team engagement — this is a controlled emulation, not an open-ended penetration test.
- Remediation engineering — the project identifies gaps and proposes playbook responses; large-scale remediation is an operational matter.

## 4.2 Four-Phase Methodology

| Phase                                          | Objective                                                                                         | Key Activities                                                                                                      | Curriculum Alignment                                                                   |
|------------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <b>Phase 1 Threat Intelligence &amp; Recon</b> | Identify and profile a real financial threat group and map its TTPs to MITRE ATT&CK.              | Research FIN7 / Lazarus Group; build ATT&CK Navigator heat map; ingest IoCs into MISP.                              | Threat Intelligence, E-Banking Security, Advisory & Emerging Technology                |
| <b>Phase 2 Red Team Emulation</b>              | Safely execute the mapped attack chain in a controlled environment to validate existing defences. | Deploy Atomic Red Team / Caldera scenarios; test access controls, OS hardening, and network segmentation.           | Penetration Testing, Network Security, OS Security, Access Control Security            |
| <b>Phase 3 Blue Team Detection Engineering</b> | Convert raw emulation telemetry into high-fidelity, low-noise SIEM detection rules.               | Analyse Sysmon & network logs; author and tune Sigma/SPL detection rules; validate against benign baselines.        | SIEM, Database Security, Vulnerability Assessment / Management                         |
| <b>Phase 4 IR Playbook &amp; Mitigation</b>    | Document a complete, actionable Tier 2/3 response playbook for the emulated threat.               | Define containment, eradication, and recovery steps; align with data classification and risk management frameworks. | Incident Response Management, Technology Risk Management, Data Classification, BCP/DRP |

## 4.3 Deliverables

1. Threat Emulation Runbook — Full documentation of the selected attack path, tools, commands, and observed outcomes. Serves as a repeatable test procedure for future exercises.
2. Custom SIEM Detection Rules — A set of three or more tested, high-fidelity alert rules (written in Sigma or platform-native query language) ready for deployment into the bank's SIEM environment.
3. Targeted IR Playbook — A step-by-step Tier 2/3 Incident Response guide specific to the emulated threat, including containment actions, forensic preservation steps, stakeholder communication templates, and post-incident review checklist.

## 5. Timeline & Milestones

The project is structured across 12 weeks, divided into the four phases. Key milestones are indicated below.

| Week  | Phase                           | Key Activities                                                                                                         | Milestone / Deliverable                                                                     |
|-------|---------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| 1–2   | Phase 1 – Threat Intelligence   | Select threat group; build MITRE ATT&CK Navigator map; ingest IoCs into MISP; document TTP profile.                    | <b>MILESTONE 1: ATT&amp;CK heat map and threat actor profile approved by supervisor.</b>    |
| 3     | Phase 1 – Wrap-up & Lab Setup   | Finalise threat profile; set up isolated lab environment; obtain approval to proceed with emulation.                   | Lab environment signed off.                                                                 |
| 4–6   | Phase 2 – Red Team Emulation    | Execute Atomic Red Team / Caldera scenarios for each mapped TTP; document observations and log evidence.               | <b>MILESTONE 2: Emulation Runbook (Draft).</b>                                              |
| 7–9   | Phase 3 – Detection Engineering | Analyse emulation telemetry; author SIEM detection rules; tune against benign baselines; validate false-positive rate. | <b>MILESTONE 3: Custom SIEM detection rules tested and documented.</b>                      |
| 10–11 | Phase 4 – IR Playbook           | Draft containment and eradication steps; align with BCP/DRP and Risk Management frameworks; peer review.               | <b>MILESTONE 4: IR Playbook (Draft) submitted for advisor review.</b>                       |
| 12    | Final Review & Presentation     | Finalise all artefacts; prepare presentation slides; deliver final report and live demo to supervisor.                 | <b>FINAL DELIVERABLE: All three artefacts delivered; internship presentation completed.</b> |

## 6. Project Resources & Budget

### 6.1 Technical Resources

| Resource                     | Purpose                                           | Source / Cost      |
|------------------------------|---------------------------------------------------|--------------------|
| MITRE ATT&CK Navigator       | Map and visualise threat actor TTPs               | Open Source — Free |
| MISP (Threat Intel Platform) | Ingest and manage Indicators of Compromise (IoCs) | Open Source — Free |
| Atomic Red Team / Caldera    | Controlled adversary technique emulation          | Open Source — Free |
| Metasploit Framework         | Supplementary exploitation framework for Phase 2  | Open Source — Free |
| Windows Sysmon               | Endpoint telemetry and process logging            | Microsoft — Free   |

|                                       |                                                        |                                  |
|---------------------------------------|--------------------------------------------------------|----------------------------------|
| <b>SIEM (Splunk / ELK / Sentinel)</b> | Log aggregation, detection rule authoring and alerting | Bank-provided or free tier       |
| <b>Isolated Lab Environment (VM)</b>  | Safe execution of emulation scenarios                  | Bank-provided or personal laptop |

## 6.2 Budget Estimate

This project is designed to be executed at minimal cost by leveraging open-source tooling exclusively. The primary resource requirements are time and access to a lab environment, both of which fall within normal internship parameters.

| Item                                                     | Estimated Cost     | Notes                               |
|----------------------------------------------------------|--------------------|-------------------------------------|
| Open-source tooling (MISP, Atomic Red Team, Sysmon, ELK) | USD 0              | All free / community editions       |
| Lab environment (bank-provided VM or personal laptop)    | USD 0              | Requires IT approval if bank-hosted |
| SIEM platform access (Splunk Free Trial or ELK)          | USD 0              | Free tiers sufficient for lab scale |
| Report printing & documentation                          | < USD 10           | Final hard-copy deliverable         |
| <b>TOTAL</b>                                             | <b>&lt; USD 10</b> | Excludes supervisor time            |

## 7. Risk Management & Sustainability

| Risk                                                      | Severity      | Likelihood                                                  | Mitigation Strategy                                                                                            |
|-----------------------------------------------------------|---------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Emulation escapes lab / impacts production systems</b> | <b>HIGH</b>   | Low — requires strict VM isolation and network segmentation | Obtain written IT approval; air-gap lab environment; all tests reviewed by supervisor before execution.        |
| <b>Scope creep leads to incomplete deliverables</b>       | <b>MEDIUM</b> | Medium — complex multi-tool project                         | Enforce single attack-vector constraint; weekly check-ins with supervisor; use phase milestones as hard gates. |
| <b>SIEM access not granted in time</b>                    | <b>MEDIUM</b> | Low–Medium                                                  | Begin Phase 3 with ELK Stack locally; migrate rules to bank SIEM if/when access is provided.                   |
| <b>Insufficient log telemetry from emulation</b>          | <b>MEDIUM</b> | Low — Sysmon provides rich endpoint telemetry               | Pre-configure Sysmon with SwiftOnSecurity baseline before Phase 2; validate logging before emulation begins.   |



|                                    |     |        |                                                                                                             |
|------------------------------------|-----|--------|-------------------------------------------------------------------------------------------------------------|
| Tool/platform compatibility issues | LOW | Medium | Standardise on open-source, platform-agnostic Sigma rule format; document conversion steps for target SIEM. |
|------------------------------------|-----|--------|-------------------------------------------------------------------------------------------------------------|

## 7.1 Sustainability

All artefacts produced by this project are designed for ongoing operational use beyond the internship period. The Threat Emulation Runbook serves as a repeatable procedure for future purple team exercises. The SIEM detection rules are written in a platform-agnostic format (Sigma) to allow migration if the bank changes tooling. The IR Playbook follows a modular structure so that individual sections can be updated as the threat landscape evolves without requiring a complete rewrite.

## 8. Evaluation Metrics (Monitoring)

Success will be assessed against the following Key Performance Indicators (KPIs) throughout and at the conclusion of the project:

| Phase   | KPI                                                              | Target                            | Measurement Method                    |
|---------|------------------------------------------------------------------|-----------------------------------|---------------------------------------|
| Phase 1 | Number of ATT&CK techniques documented for selected threat group | 10+ techniques mapped             | ATT&CK Navigator export               |
| Phase 2 | Percentage of mapped TTPs successfully emulated in lab           | >= 80%                            | Emulation Runbook logs                |
| Phase 2 | Existing defences bypassed vs. blocked (detection gap baseline)  | Documented baseline established   | Comparison with Phase 3 results       |
| Phase 3 | Number of custom SIEM rules authored and validated               | >= 3 high-fidelity rules          | SIEM rule export + test evidence      |
| Phase 3 | False-positive rate of new detection rules in lab                | < 5%                              | Rule testing against benign event log |
| Phase 4 | IR Playbook completeness (all attack stages covered)             | 100% of emulated stages addressed | Supervisor review checklist           |
| Overall | All three artefacts delivered before final presentation          | 100% on time                      | Project milestone log                 |

## 9. Conclusion & Next Steps

The Intelligence-Driven Purple Team Pipeline represents an ambitious but achievable project that is uniquely suited to both the internship curriculum and the bank's real-world security needs. By

grounding every phase in a verified, real-world threat group and producing three immediately deployable artefacts, this project goes beyond academic exercise — it makes a direct, measurable contribution to the bank's defensive capability.

The project also serves as a clear demonstration of Tier 2/3 analyst competency: the ability to think like an attacker, engineer detections like a defender, and communicate risk like a security professional. These are exactly the skills required for a career trajectory toward Threat Hunting and Advanced Incident Response.

### 9.1 Requested Next Steps

- 1. Advisor review and feedback on this proposal during our next scheduled discussion.
- 2. Agreement on the specific threat group and attack vector to be used in Phase 1.
- 3. Initiation of the IT approval process for the isolated lab environment.
- 4. Approval to begin Phase 1 immediately upon confirmation of the above.

**Request: Approval to initiate Phase 1 of the project upon completion of our next advisory discussion.**

## 10. Appendices

### Appendix A — Curriculum Alignment Matrix

The table below confirms that all 23 courses in the approved internship curriculum are addressed across the four project phases.

| Course Title                          | Phase 1 | Phase 2 | Phase 3 | Phase 4 |
|---------------------------------------|---------|---------|---------|---------|
| General IS Awareness                  |         |         |         | ✓       |
| Technology Risk Management Guidelines |         |         |         | ✓       |
| ISO 27001                             |         |         | ✓       | ✓       |
| ISO 22301                             |         |         |         | ✓       |
| PCI-DSS                               |         |         | ✓       | ✓       |

|                                     |   |   |   |   |
|-------------------------------------|---|---|---|---|
| Third Party Risk                    | ✓ |   |   |   |
| BCP/DRP                             |   |   |   | ✓ |
| Advisory and Emerging Technology    | ✓ |   |   |   |
| IT Risk Management                  |   |   | ✓ | ✓ |
| Database Security                   |   | ✓ | ✓ |   |
| Network Security                    |   | ✓ | ✓ |   |
| Access Control Security             |   | ✓ |   |   |
| Asset Inventory Management          | ✓ |   |   |   |
| OS Security                         |   | ✓ |   |   |
| Data Classification                 |   |   |   | ✓ |
| Data Protection                     |   |   | ✓ | ✓ |
| Data Privacy                        |   |   |   | ✓ |
| SIEM                                |   |   | ✓ |   |
| Vulnerability Assessment/Management |   | ✓ | ✓ |   |
| Incident Response Management        |   |   |   | ✓ |
| Threat Intelligence                 | ✓ |   |   |   |
| Penetration Testing                 |   | ✓ |   |   |
| E-Banking Security                  | ✓ | ✓ | ✓ | ✓ |

## Appendix B — Candidate Threat Groups

Two primary threat groups are proposed for supervisor selection. Final choice will be confirmed during the Phase 1 discussion.

| Group                | Attribution         | Primary TTPs                                                                        | Why Relevant                                                                                           |
|----------------------|---------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Lazarus Group</b> | Nation-State (DPRK) | Spear-phishing, WannaCry-style lateral movement, SWIFT fraud, database exfiltration | Directly responsible for multiple high-profile banking heists; extensively documented in MITRE ATT&CK. |
| <b>FIN7</b>          | Criminal Syndicate  | Spear-phishing, Carbanak backdoor, POS/payment system targeting, credential theft   | Financially motivated; well-documented attack chain against banking and retail financial targets.      |

## Appendix C — Glossary

**APT:** Advanced Persistent Threat — a sophisticated, often state-sponsored threat actor conducting long-term targeted attacks.

**ATT&CK:** MITRE's Adversarial Tactics, Techniques & Common Knowledge framework — the industry-standard taxonomy for threat behaviour.

**BCP/DRP:** Business Continuity Plan / Disaster Recovery Plan.

**CTI:** Cyber Threat Intelligence — structured information about threat actors, their capabilities, and their intentions.

**IoC:** Indicator of Compromise — artefacts (IPs, hashes, domains) that indicate a system may have been compromised.

**IR Playbook:** Incident Response Playbook — a documented, step-by-step guide for responding to a specific type of security incident.

**MISP:** Malware Information Sharing Platform — an open-source threat intelligence platform.

**Purple Team:** A collaborative security exercise where Red Team (attack) and Blue Team (defence) work together to improve detection.

**SIEM:** Security Information and Event Management — platform for aggregating, correlating, and alerting on security events.

**Sigma:** An open, vendor-agnostic format for writing SIEM detection rules, convertible to platform-specific query languages.

**SOC:** Security Operations Centre — the team responsible for monitoring, detecting, and responding to security events.

**Sysmon:** System Monitor — a Windows system service that logs detailed process creation and network activity for forensic analysis.

**TTP:** Tactics, Techniques & Procedures — the behavioural fingerprint of a threat actor.